

Corridors, Not Detours: Bounding Language Preferences by Local Opportunity and Embodiment

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Abstract—We present ongoing work on bounding language-conditioned outdoor route preferences by the alternatives available near a start–goal query and by the active robot’s capabilities. Our central proposal, *corridor-feasible language alignment*, separates transferable mission values from environment- and embodiment-dependent route consequences. A semantic region graph exposes bounded alternatives. Preferences without meaningful local opportunity are reported as unsupported rather than inducing arbitrary detours. Robot-specific Jackal and Spot cost models are trained using simulated traversal data. In held-out simulated terrain scenarios with typed intents, the proposed planner changes the route in the requested direction in 94.6% of atomic cases where a meaningful local alternative exists. Against neutral routing, the selected routes recover most of the preference improvement available within each corridor, while every candidate respects the simulated robot-cost budget. These preliminary results demonstrate corridor-bounded route adaptation in simulation. Free-form language grounding and physical robot validation remain future work.

I. INTRODUCTION

Outdoor operators naturally describe route preferences such as *avoid severe slopes* or *preserve energy*. Yet an instruction alone does not determine a useful route: the requested alternative may not exist near the start–goal query, and different robots may experience the same terrain differently.

For example, an operator might say: “*Prefer smooth, high-traction terrain while keeping robot risk low.*” Smoothness and traction express mission values. Their risk, energy, and mobility consequences depend on the robot. Within the same local opportunity, a legged robot may accept a steeper direct segment while a wheeled robot selects a longer, firmer one. If the feasible alternatives do not differ meaningfully in the requested terrain, the planner should report the preference as unsupported rather than take an unrelated detour. This is the intended free-form interface. The current evaluation uses typed physical intents and does not test unrestricted language understanding.

Language navigation has grounded destinations, landmarks, actions, and route instructions [1], [2], [3], [4]. Terrain systems additionally estimate traversability and language-conditioned cost maps [5], [6], [7]. We ask a complementary question: *when should a preference change a feasible route, and how should that response depend on the robot?*

Our hypothesis is that transferable mission values should be separated from environment- and embodiment-dependent consequences. We propose a corridor contract that bounds where preferences may act and makes lack of opportunity explicit. The cross-domain lesson is to transfer the requested

value, not the route itself: opportunity and traversal consequences remain local to the map and embodiment. Figure 1 summarizes the framework.

II. PROPOSED DIRECTION

Given map M , robot r , instruction u , and route π , the planner separates language-derived values from robot-conditioned route consequences (e.g., energy, risk, and terrain exposure). This factorization is related to successor features and language grounding through robot affordances [8], [9], and is used here to swap robot models without changing the mission vocabulary.

The proposed end-to-end realization (Figure 1) encodes language, terrain, and robot context jointly; the current evaluation begins from typed intents and does not test the free-form language encoder.

Planning is conditioned on *corridor opportunity*. A method-independent corridor $\mathcal{C}$ is fixed from robot, distance, and available objective-specific lower-bound routes before the evaluated route is generated. Let $\mathcal{F}_r(\mathcal{C})$ be the routes inside $\mathcal{C}$ that satisfy robot r ’s mobility constraints, and let $J_r(\pi)$ be the robot-specific traversal cost of route π . The admissible routes remain within a relative margin δ of the minimum feasible cost J_r^* :

$$\begin{aligned} J_r^* &= \min_{\pi \in \mathcal{F}_r(\mathcal{C})} J_r(\pi), \\ \Pi_{\mathcal{C},\delta} &= \{\pi \in \mathcal{F}_r(\mathcal{C}) : \\ &\quad J_r(\pi) \leq (1 + \delta)J_r^*\}. \end{aligned} \quad (1)$$

Prior work evaluates language-conditioned routes through preferred-terrain exposure and rank regret [6], [7]. Following these route-level tests, we evaluate response only when feasible alternatives differ meaningfully in the requested property. A response succeeds when the selected route shifts a nontrivial amount from the neutral route in the requested direction. For compound requests, we measure each requested property separately and also whether all properties improve together.

Motivated by regret-based comparison of alternatives in multiobjective robot planning [10], we compare the neutral route π_0 with a reference π_j^* representing an approximate objective-specific choice from the same feasible alternatives.

Corridor-Bounded Intent Regret (CBIR) places the reference at 0 and the neutral route at 1. For a minimization-form objective g_j ,

$$\text{CBIR}_j(\hat{\pi}) = \frac{g_j(\hat{\pi}) - g_j(\pi_j^*)}{g_j(\pi_0) - g_j(\pi_j^*)}. \quad (2)$$

We apply CBIR only when the reference improves on

Language-conditioned two-stage local-global attention

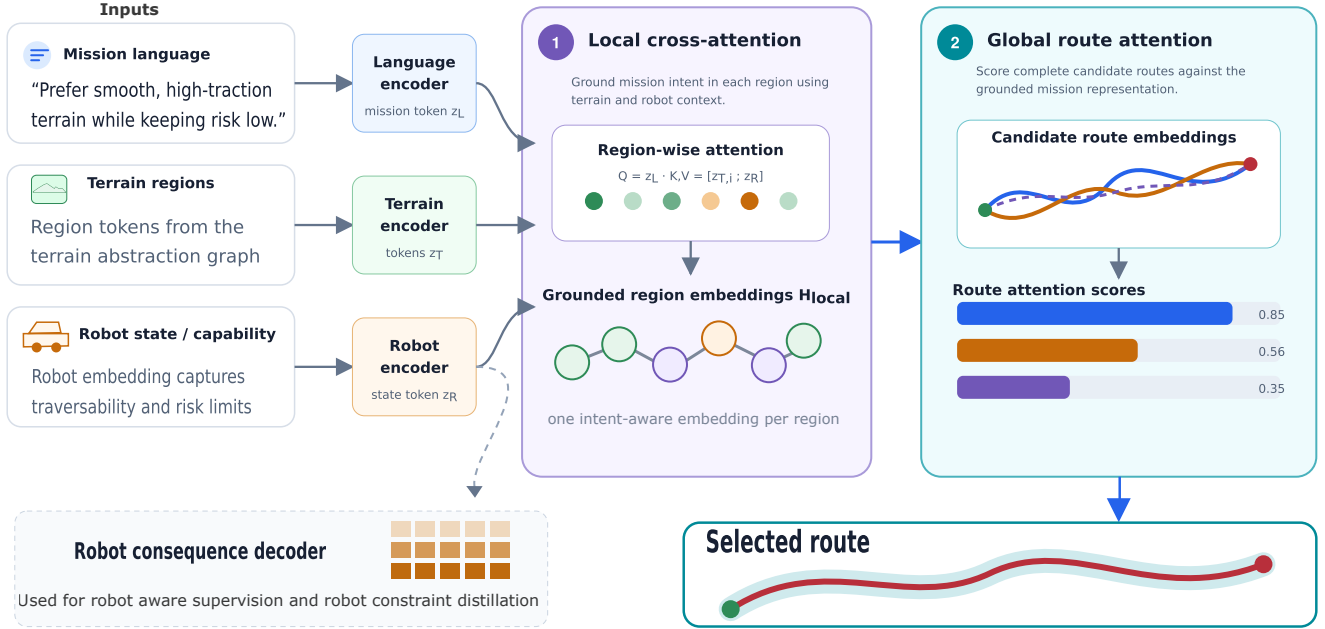


Fig. 1. Proposed two-stage framework. Local cross-attention relates language values to terrain and robot context. Global attention then scores bounded routes before refinement. The robot branch enables robot-aware cost prediction.

Terrain intent: "Prefer forested terrain while remaining robot-feasible."

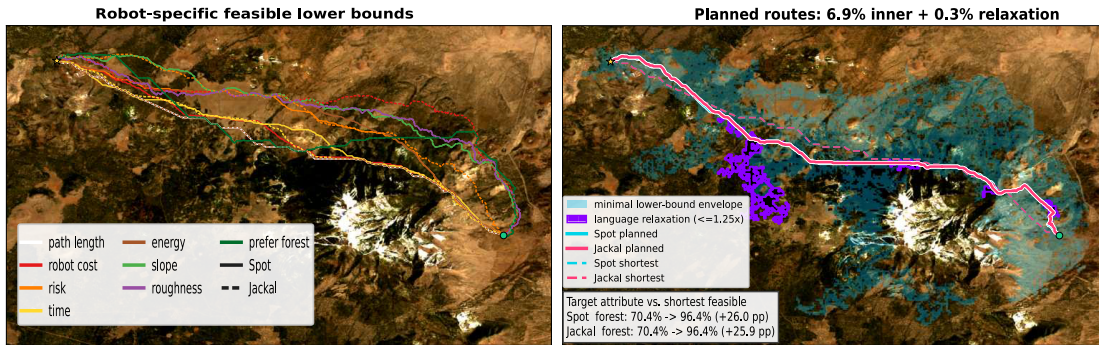


Fig. 2. A held-out simulated terrain request. Independent lower bounds define the corridor (left), and both robots increase forest exposure while remaining near their feasible length lower bounds (right).

Agent intent: "Minimize traversal risk for the active robot."

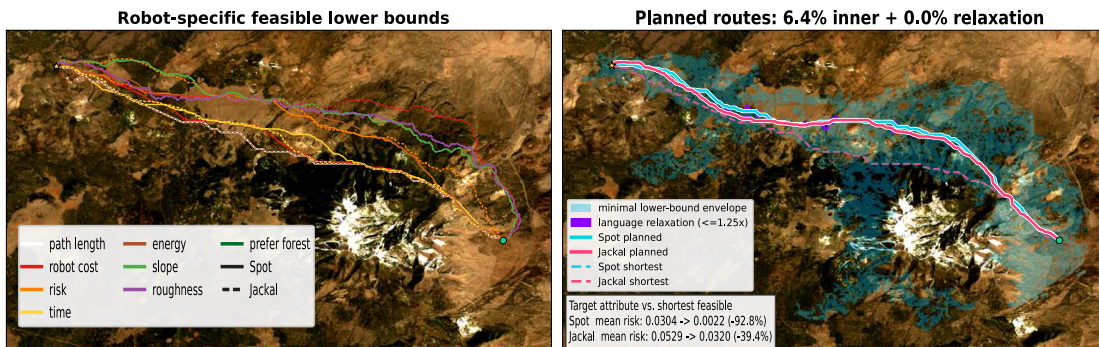


Fig. 3. A simulated robot-conditioned risk request on the same query. Spot and Jackal select different responses within their robot-specific corridors.

the neutral route. Because the reference is approximate, we interpret CBIR together with corridor containment and robot-cost compliance.

An abstract graph is needed to bind language preferences and robot consequences to route-scale terrain units while exposing their connectivity for candidate search. We use the CLEAR semantic–geometric abstraction, which converts large terrain maps into connected, semantically aligned, elevation-aware regions [11]. The prototype predicts robot-specific consequences on this graph, then generates, selects, and refines bounded route alternatives.

III. INITIAL RESULTS

We evaluated the proposed planner on held-out simulated terrain scenarios with separate Jackal and Spot cost models. The study covers typed slope, roughness, and traction intents. It does not yet evaluate free-form language or physical traversal.

Across single-property requests with meaningful alternatives, the planner changes the route in the requested direction in 94.6% of cases. Mean CBIR is 0.146 (lower is better, with 0 denoting the approximate reference route and 1 the neutral route), indicating most of the available preference improvement is recovered. Every selected route satisfies the simulated robot-cost budget. Compound requests are less reliable, especially when all requested properties must improve simultaneously. Figures 2 and 3 illustrate how terrain and robot consequences alter routes within the same bounded scope. These diagnostics establish bounded directional response, not human-preference accuracy, physical calibration, or superiority to other planning methods.

IV. VALIDATION AGENDA

The next evaluation focuses on four tests:

- **Generalization:** independent maps and sensor conditions, including cases where no meaningful alternative exists.
- **Embodiment:** paired Jackal–Spot trials using the same maps and instructions while measuring physical outcomes.
- **Alignment:** blinded route comparisons against human preferences, including disagreement and inter-rater reliability.
- **Controls:** matched comparisons with adaptive search and existing language-conditioned cost-map methods [6], [7].

V. CONCLUSION

Corridor feasibility provides a compact contract between transferable language values, local opportunity, and embodiment-specific traversal. The early typed-intent simulation supports this contract as a measurable research direction. Independent topologies, human judgments, matched controls, and paired physical trials are the decisive next tests.

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